The University of Texas at Austin
Optimization

HOMEWORK 5 SOLUTIONS

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1. **Linear Algebra**

   (a) Compute the eigenvalues and eigenvectors of the matrix

   $$A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

   and show that the eigenvectors are orthogonal to each other.

   Solution: *The eigenvalues are* $\lambda = 1, 2$. *The eigenvectors are*

   $$\boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \boldsymbol{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

   Since $\boldsymbol{v}_1^\top \boldsymbol{v}_2 = 0$, *these are orthogonal.*

   (b) Compute the eigenvalues and eigenvectors of the matrix

   $$A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$$

   and show that the eigenvectors are not orthogonal to each other.

   Solution: *The eigenvalues are* $\lambda = 1, 2$. *The eigenvectors are*

   $$\boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \boldsymbol{v}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

   Note that by convention, eigenvalues are given as normalized vectors, i.e., with Euclidean norm equal to one. Since $\boldsymbol{v}_1^\top \boldsymbol{v}_2 \neq 0$, *these are not orthogonal. Thus this is consistent with the theory: symmetric matrices have orthogonal eigenvectors, where as non-symmetric matrices do not.*

   (c) Compute the eigenvalues and eigenvectors of the matrix

   $$A_3 = \begin{bmatrix} 4 & 11 & 11 & 12 \\ 11 & 13 & 14 & 12 \\ 11 & 14 & 16 & 16 \\ 12 & 12 & 16 & 18 \end{bmatrix}.$$

   Is this matrix positive semidefinite? Are the eigenvectors orthogonal to each other?

   Solution: *This is easy to do, e.g., using Python.*

```
// Hello.python
import numpy as np
// Given matrix Q
[values,vectors]=np.linalg.eig(Q)
// Check whether eigenvectors are orthogonal.
np.dot(vectors[:,1],e[:,3])
```

(d) Repeat this with random matrices of your choosing. Note that symmetric matrices, whether positive definite or note (i.e., whether or not they have nonnegative eigenvalues) will always have eigenvectors that are perpendicular (orthogonal) to each other.

2. Given any functions $f_1(x), f_2(x), ..., f_n(x)$, prove that the following are equivalent.

(a)

$$\min_x \sum_{i=1}^{k} f_{[i]}(x),$$

where $f_{[i]}(x)$ is the ith largest of the set of values $f_1(x), f_2(x), ..., f_n(x)$. (Note that which f_i are the largest may change as x changes.)

(b)

$$\begin{aligned} \min_{x,y,t} \quad & kt + \sum_{i=1}^{n} y_i \\ \text{subject to} \quad & f_j(x) \leq t + y_j \qquad \forall j = 1, ..., n \\ & y \geq 0. \end{aligned}$$

Solution *Suppose you get an optimal solution x^* from (a). You can find a feasible point in (b) with $(x,t) = (x^*, f_{[k]}(x^*))$ and setting y_j as*

$$\begin{aligned} y_j = f_{[j]}(x^*) - f_{[k]}(x^*) \qquad & j \leq k, \\ y_j = 0 \qquad & \textit{otherwise} \end{aligned} \tag{1}$$

Then, (b) has same objective value with feasible point. That is, $p_b^* \leq p_a^*$ where p_a^* is an optimal value of (a) and p_b^* is that of (b).

On the other side, if x is a feasible point for (b), then minimum possible value with varying t and y is the objective function in (a), which is $\min_x \sum_{i=1}^{k} f_{[i]}(x)$. Otherwise, you can always decrease the objective value. Therefore, if x^* is an optimal solution, it is the feasible point in (a) with same objective value. That is, $p_a^* \leq p_b^*$. Thus, $p_a^* = p_b^*$.

3. Consider the LP from Homework 3:

$$\begin{aligned} \min: \quad & x_1 - x_2 \\ \text{s.t.}: \quad & 2x_1 + 3x_2 - x_3 + x_4 \leq 0 \\ & 3x_1 + x_2 + 4x_3 - 2x_4 \geq 3 \\ & -x_1 - x_2 + 2x_3 + x_4 = 6 \\ & x_1 \leq 0 \\ & x_2, x_3 \geq 0. \end{aligned}$$

Write this explicitly as an SDP. Thus, your objective must be linear in the decision variables you choose, and your constraints can have equality constraints and semidefinite constraints, but not coordinate-wise inequality constraints as we have in LP.

Solution: *We follow the idea set out in the lecture: if we have only diagonal matrices, then indeed the SDP is equivalent to an LP, since a diagonal matrix is positive semidefinite if and only if the vector of its diagonal is component-wise nonnegative. Thus we define diagonal matrices:*

$$\begin{aligned} A_0 &= \text{diag}(0, -3, -6, 6, 0, 0, 0) \\ A_1 &= \text{diag}(-2, 3, -1, 1, -1, 0, 0) \\ A_2 &= \text{diag}(-3, 1, -1, 1, 0, 1, 0) \\ A_3 &= \text{diag}(1, 4, 2, -2, 0, 0, 1) \\ A_4 &= \text{diag}(-1, -2, 1, -1, 0, 0, 0). \end{aligned}$$

Then the equivalent SDP becomes:

$$\begin{aligned} \min: \quad & x_1 - x_2 \\ \text{s.t.}: \quad & A_0 + A_1 x_1 + A_2 x_2 + A_3 x_3 + A_4 x_4 \succeq 0 \end{aligned}$$

4. Consider the following non-linear optimization problem:

$$\begin{aligned} \min: \quad & \frac{(\boldsymbol{c}^\top \boldsymbol{x})^2}{\boldsymbol{d}^\top \boldsymbol{x}} \\ \text{s.t.}: \quad & A\boldsymbol{x} \geq \boldsymbol{b}, \end{aligned}$$

where we assume that $A\boldsymbol{x} \geq \boldsymbol{b} \Rightarrow \boldsymbol{d}^\top \boldsymbol{x} > 0$.

(a) Show that this problem is convex.

Solution: *This is implied by the next part.*

(b) Reformulate this problem as an SDP. *(Hint: it may be useful for you to add an additional dummy variable that will allow you to move the objective function into the constraints).*

Solution: *As suggested by the hint, we first transform the problem by adding a dummy variable, so as to move the fractional objective to the constraints:*

$$\begin{aligned} \min: \quad & \gamma \\ \text{s.t.}: \quad & A\boldsymbol{x} \geq \boldsymbol{b} \\ & \frac{(\boldsymbol{c}^\top \boldsymbol{x})^2}{\boldsymbol{d}^\top \boldsymbol{x}} \leq \gamma. \end{aligned}$$

Now note that the non-linear constraint

$$\frac{(\boldsymbol{c}^\top \boldsymbol{x})^2}{\boldsymbol{d}^\top \boldsymbol{x}} \leq \gamma,$$

can be expressed as the semidefinite constraint

$$\begin{bmatrix} \gamma & \boldsymbol{c}^\top \boldsymbol{x} \\ \boldsymbol{c}^\top \boldsymbol{x} & \boldsymbol{d}^\top \boldsymbol{x} \end{bmatrix} \succeq 0.$$

Then, again using the same reasoning as we did in the previous exercise to convert the linear inequalities into semidefinite constraints, we have:

$$\begin{array}{ll} \min: & \gamma \\ \text{s.t.}: & \begin{bmatrix} \operatorname{diag}(A\boldsymbol{x} - \boldsymbol{b}) & 0 & 0 \\ 0 & \gamma & \boldsymbol{c}^\top \boldsymbol{x} \\ 0 & \boldsymbol{c}^\top \boldsymbol{x} & \boldsymbol{d}^\top \boldsymbol{x} \end{bmatrix} \succeq 0. \end{array}$$

5. **Using KKT**.

Consider the problem we saw in the lecture:

$$\begin{array}{ll} \min: & x_1^2 - 6x_1 + x_2^2 - 4x_2 \\ \text{s.t.}: & x_1^2 - 2x_1 + x_2^2 \leq 0 \\ & \frac{1}{2}x_1^2 - 2x_1 - x_2 + 2 \leq 0 \\ & x_1 - x_2 \leq 0 \end{array}$$

(a) Plot the constraints and shade the feasible region.

Solution: *See Figure 5a. We show all constraints, along with the feasible set (shaded). Also, the circles are the level sets of the objective function. Thus, it is graphically easy to verify the optimal solution. However, we do not want to relegate ourselves to graphical solutions, since this would mean only being able to solve two or at most three dimensional problems.*

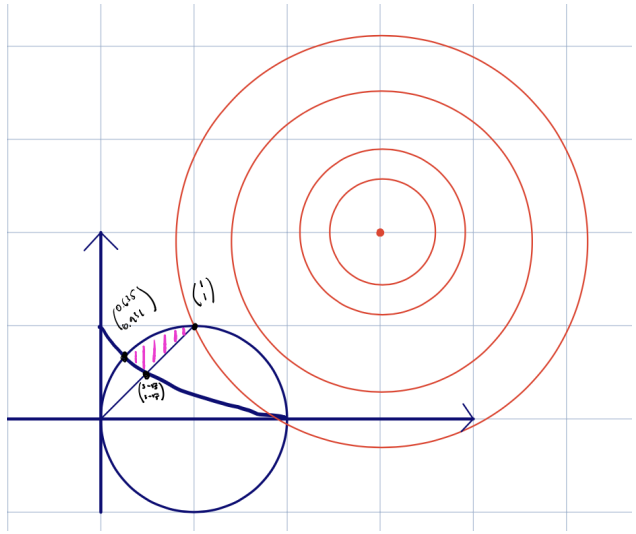
(b) Verify that the point $\boldsymbol{x} = (1, 0)^\top$ is an optimal solution, using the KKT equations. We did part of this in class. We did not show that complementary slackness holds.

Solution: *With what we did in lecture, this is straightforward.*

(c) Compute the other two corner points (the intersection of constraints 1 and 2, and then the intersection of constraints 1 and 3). Verify, using KKT, that neither of these is an optimal solution.

Solution: *The other two points are obtained by solving the corresponding equality constraints. The values are as given in the graph. Note that for the point* $(0.635, 0.935)$ *the actual expression is quite complicated, so this is just a three decimal approximation.*

To show that these points are not optimal, we have to show that it is not possible to satisfy the KKT conditions there, namely, dual feasibility, complementary slackness and



the stationarity condition (note that primal feasibility is already established). We do this for the point $(0.635, 0.935)$*. For the other point, the method is identical.*

Complementary slackness together with dual feasibility and stationarity read: $\exists \lambda_1, \lambda_2 \geq 0$ *such that:*

$$\nabla f_0(\boldsymbol{x}) + \lambda_1 \nabla f_1(\boldsymbol{x}) + \lambda_2 \nabla f_2(\boldsymbol{x}) = 0,$$

where $\boldsymbol{x}^\top = (0.635, 0.935)$*, and* $f_0(\cdot)$ *is our objective function,* $f_1(\cdot)$ *is the first constraint, and* $f_2(\cdot)$ *the second constraint. Note that by complementary slackness,* $\lambda_3 = 0$*, and hence the third constraint does not appear in the stationarity condition. Computing these out, we get two equations, which will allow us to solve for the two unknowns,* λ_1 *and* λ_2*:*

$$\begin{aligned} 2x_1 - 6 + \lambda_1(2x_1 - 2) + \lambda_2(x_1 - 2) &= 0 \\ 2x_2 - 4 + \lambda_1(2x_2) - \lambda_2 &= 0. \end{aligned}$$

Solving now for λ_2 *in terms of* λ_1*, plugging into the first equation, and solving for* λ_1*, we find:*

$$\lambda_1 = \frac{6 - 2x_1 - (x_1 - 2)(2x_2 - 4)}{2x_1 - 2 + (x_1 - 2)2x_2},$$

but evaluating this (again recall that x_1 *and* x_2 *are fixed) gives us a negative value, hence violating dual feasibility.*

(d) We have thus checked that of the three corner points, one is an optimal solution, and the other two are not. There are still points in the interior, that could potentially be optimal solutions. Use the KKT conditions to show that this cannot be the case, i.e., for any interior point $\tilde{\boldsymbol{x}}$, that is, any point for which the three constraints are *strictly* satisfied, $\tilde{\boldsymbol{x}}$ is not an optimal solution.

Solution: *This again follows just as the above. By complementary slackness, we now must have* $\lambda_1 = \lambda_2 = \lambda_3 = 0$*, and hence the stationarity condition reads:*

$$\nabla f_0(\boldsymbol{x}) = 0,$$

but this is only satisfied at the point $(3, 2)$*, and hence we satisfy dual feasibility, complementary slackness and stationarity, but violate primal feasibility.*

6. Consider the optimization problem:

$$\begin{aligned} \min: \quad & \frac{1}{2}\boldsymbol{x}^\top Q\boldsymbol{x} + \boldsymbol{q}^\top \boldsymbol{x} + c_0 \\ \text{s.t.}: \quad & \boldsymbol{x} \in [-1,1]^3, \end{aligned}$$

where the data of the problem are given as:

$$Q = \begin{bmatrix} 13 & 12 & -2 \\ 12 & 17 & 6 \\ -2 & 6 & 12 \end{bmatrix}, \qquad \boldsymbol{q} = \begin{pmatrix} -22 \\ -14.5 \\ 13 \end{pmatrix}, \qquad c_0 = -1.$$

(a) Is this a convex optimization problem? Explain your answer precisely.

Solution: *As you can check in a number of ways (e.g., Python), the matrix Q has nonnegative (in fact, strictly positive) eigenvalues, and hence the objective is convex, as we showed in lecture, since it is the sum of a convex function, a linear function (which is always convex) and a scalar (which is always convex). The constraint set is the cube in 3 dimensions, and hence also convex.*

(b) Show that $\hat{\boldsymbol{x}}^\top = (1, 0.5, -1)$ is an optimal solution.

Solution: *Again we can do this using the KKT conditions. The point is clearly primal feasible. Thus we need to find a dual feasible solution that satisfies complementary slackness, and also satisfies the stationarity condition. There are six constraints:*

$$x_i \leq 1, \quad -x_i \leq 1, \quad i = 1, 2, 3.$$

Compementary slackness thus means only the dual variables corresponding to the tight constraints can be non-zero. Now the stationarity condition reads:

$$Q\hat{\boldsymbol{x}} + \boldsymbol{q} + \lambda_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Plugging in the values for Q, $\boldsymbol{q}$ and $\hat{\boldsymbol{x}}$, we find that setting $\lambda = 1$ and $\lambda_2 = 0.5$, satisfies the stationarity condition. Since these values are nonnegative, dual feasibility are also satisfied, and hence the KKT conditions are satisfied. The point $\hat{\boldsymbol{x}}^\top = (1, 0.5, -1)$ is an optimal solution.

(c) Are there other optimal solutions?

Solution: *We could go through all possibilities and repeat the above exercise, to find that there are none. A more direct way is to note that the eigenvalues of Q are strictly positive, and hence the objective function is strongly convex, and hence must have a unique optimal solution.*